PHAN CANH TUAN DAT
Full-Stack Developer

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Portfolio: https://ttuandatt.github.io/AWS-Workshop/

SUMMARY

Full-stack developer who builds products end-to-end. Experienced with Next.js (App Router), NestJS, TypeScript, PostgreSQL, and Redis. Built an AI-powered IELTS platform with async scoring pipeline (BullMQ, OpenAI/Gemini), real-time feedback via SSE, and role-based dashboards (Learner/Instructor/Admin). Currently shipping a GPS food-tourism system with React Native (Expo), NestJS, and AI-driven TTS. Completed a 6-month AWS internship building Spring Boot microservices with Kafka, Redis, and CI/CD on ECS Fargate. IELTS 7.0. Passionate about shipping fast, solving real user problems, and iterating based on feedback.

PERSONAL EXPERIENCE

1. **AWS First Cloud Journey Internship** **Jun 2025 – Dec 2025**
   - Project: SaaS Task Management Web Application
   - Team size: 4 members
   - GitHub: https://ttuandatt.github.io/AWS-Workshop/
   - Description: A Todoist-like SaaS platform with 6 microservices (Auth, User, Taskflow, Notification, API Gateway, AI Model) handling project management, real-time collaboration, RBAC, and AI-powered task labeling. Deployed on AWS ECS Fargate with CI/CD via GitHub Actions.
   - Frameworks & Libraries: Spring Boot 3.4, Spring Security 6, Spring Cloud Gateway, Spring Data Redis, Spring Kafka, MapStruct, Swagger/OpenAPI
   - Technology Stack:
     - **Backend:** Spring Boot 3.4 microservices (6 services), Java 21, Spring Security 6
     - **Database:** MySQL 8.0 / Aurora MySQL, Redis 7 (cache, pub/sub, rate limiting)
     - **Messaging:** Apache Kafka 3.5 (async event-driven communication between services)
     - **Frontend:** React 19, Redux Toolkit, TailwindCSS, Axios
     - **Containerization:** Docker, Docker Compose, AWS ECS Fargate, ECR, GitHub Actions CI/CD
   - Responsibilities:

- Designed and implemented Auth Service with JWT (HMAC-SHA256), OAuth2 Google login, OTP email verification, and Redis-backed token blacklisting.
- Built API Gateway integration (Spring Cloud Gateway) with JWT filter for request routing and user context propagation.
- Implemented CI/CD pipeline with GitHub Actions for automated build, push to ECR, and deploy to ECS Fargate — reducing manual release time by 50%.
- Designed AWS cloud architecture: ECS Fargate, RDS Aurora, ElastiCache Redis, S3, ALB, Route 53, CloudWatch.
- Containerized all services using Docker and Docker Compose for consistent local and cloud environments.
- Built React frontend with Redux Toolkit for state management, TailwindCSS styling, and Axios API client.
- Collaborated with 3 teammates to decompose monolith into 6 independent microservices with clear API contracts.

2. **IELTS Helper — AI-Powered Preparation Platform** **Jan 2026 – Present**

- Project: Personal Full-Stack Web Application
- Description: A comprehensive IELTS preparation platform with AI-powered essay scoring (4-criteria IELTS rubric, <5 min turnaround), auto-graded Reading with instant explanations, classroom management, and role-based dashboards for Learners, Instructors, and Admins.
- Technology Stack:
  - **Backend:** NestJS 10, TypeScript, Prisma 6 ORM, JWT + RBAC guards, BullMQ v5 job queue
  - **Frontend:** Next.js 16 (App Router), TypeScript, TailwindCSS 4, TanStack React Query v5
  - **AI/LLM:** OpenAI GPT-4o (primary) + Google Gemini (fallback), structured JSON scoring
  - **Database:** PostgreSQL 15, Redis 7 (job queue, caching, SSE pub/sub)
  - **Real-Time:** SSE (Server-Sent Events) for live score delivery, Redis pub/sub
- Responsibilities:
  - Architected and built full-stack application from scratch: NestJS backend + Next.js frontend with 14-table PostgreSQL schema.
  - Implemented async AI scoring pipeline using BullMQ with priority queue, 3-retry exponential backoff, and dual LLM fallback (OpenAI → Gemini).
  - Built real-time score delivery via SSE (Server-Sent Events) with Redis pub/sub for instant client updates.
  - Developed auto-grading engine for 6 IELTS Reading question types with per-question explanations.
  - Implemented JWT authentication with silent token refresh, RBAC (Learner/Instructor/Admin), and Axios interceptor.
  - Created classroom module with invite codes, hierarchical content (Topics > Lessons), and instructor analytics dashboard.

3. **GPS Tours — Food Tourism Audio Guide System Jan 2026 – Present**
   - Project: Fod Tourism Audio Guide
   - Role: Lead Developer (Team size: 4)
   - GitHub: https://github.com/Ttuandatt/Seminar
   - Description: A GPS-based food tourism audio guide for Pho Vinh Khanh, District 4, HCMC. Tourists explore Points of Interest on an interactive map with AI-generated audio descriptions, real-time GPS navigation, offline caching, and multi-language support (Vietnamese/English).
   - Technology Stack:
     - **Backend:** NestJS 11, TypeScript, Prisma ORM, JWT + Passport.js, Swagger/OpenAPI
     - **Frontend:** React 19 (Vite), TailwindCSS 4, TanStack React Query, React Hook Form + Zod
     - **Mobile:** React Native (Expo SDK 54), expo-router, react-native-maps, expo-sqlite
     - **Database:** PostgreSQL 15, Redis 7 (caching), Docker Compose
     - **AI/Media:** Microsoft Edge TTS (multi-voice, multi-language audio generation)
   - Responsibilities:
     - Architected monorepo (pnpm workspaces) with NestJS API, React admin dashboard, and Expo mobile app sharing TypeScript types.
     - Built JWT authentication system with access/refresh tokens, token revocation, and role-based guards (Admin/ShopOwner/Tourist).
     - Implemented AI-driven TTS module generating multi-language audio (Vietnamese/English) for 50+ Points of Interest using Edge TTS.
     - Developed interactive map with real-time GPS tracking, POI markers, bottom sheet previews, and QR code scanner fallback.
     - Built admin dashboard with CRUD management for POIs, Tours, Merchants, and analytics KPI panels using React + TanStack Query.
     - Designed offline-first mobile architecture with expo-sqlite caching and async data sync.

EDUCATION

- Bachelor of Science in Information Technology - Sai Gon University **Sep 2022 – Dec 2026**

TECHNICAL SKILLS

- Languages: TypeScript, JavaScript, Java, SQL, HTML/CSS
- Core Concepts: Full-Stack Product Development, Microservices Architecture, RESTful API Design, Database Schema Design, Event-Driven Architecture, Async Job Queues, RBAC (Role-Based Access Control)
- Backend: NestJS, Spring Boot, Prisma ORM, Hibernate/JPA, BullMQ, Passport.js, Spring Security, Spring Cloud Gateway
- Databases: PostgreSQL, MySQL, Redis (caching, pub/sub, job queue), SQL Server
- Cloud & DevOps: AWS (ECS Fargate, ECR, RDS, S3, ALB, VPC, IAM, Route 53, CloudWatch), Docker, Docker Compose, GitHub Actions CI/CD, Vercel
- Version Control & Collaboration: Git, GitHub, Figma
- Frontend: Next.js (App Router), React, React Native (Expo), TailwindCSS, TanStack Query, Redux Toolkit, Zustand

- AI & Real-Time: OpenAI API, Google Gemini API, BullMQ async pipelines, SSE (Server-Sent Events), WebSocket (Socket.io)
- Languages: English (Advanced — IELTS 7.0), Vietnamese (Native)

CERTIFICATION

- IELTS: 7.0